CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2011

PT108 FUNDAMENTALS OF BIOMECHANICS

Date: 14.06.11, Tuesday **Time: 10.00a.m to10.20a.m** **Maximum Marks: 10**

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs) **(1 × 10 = 10)**
(Encircle the correct answer)

1. Which of the following is an example of angular motion?
   a. The ball after being thrown by the pitcher
   b. A parachutist in free fall
   c. A runner's leg motion during a 100-m race
   d. None of the above

2. There are .... planes parallel to the saggital cardinal plane.
   a. Zero
   b. Multiple
   c. Two
   d. None of the above.

3. The shaft of a long bone is called the ....
   a. Epiphysis
   b. Diaphysis
   c. Metaphysis
   d. Duraphysis

4. Synarthrodial joints allow for.........movements.
   a. Small
   b. Large
   c. Medium
   d. None of the above

5. Putting your hand in your back pocket involves..........of shoulder joint.
   a. Medial Rotation
   b. Lateral Rotation
   c. Flexion
   d. Opposition

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CHARUSAT CAMPUS CHANGA

Examination Semester ____________
Subject ________ No. ________
Candidate's Seat No. ____________
Section ____________
Date ________

Sr. Supervisor's Sign.

6. Action of Sternocleidomastoid, when acting unilaterally is ....
   a. Ipsilateral Lateral flexion & Contralateral Rotation
   b. Ipsilateral Rotation & Contralateral Lateral flexion
   c. Flexion & Contralateral Lateral flexion
   d. Flexion & Contralateral Rotation

7. A motor unit is defined as ....
   a. A single neuron and all the muscle cells it innervates
   b. A single muscle fiber and the neuron that it innervates
   c. All of the muscles in a motor pool
   d. The motor neurons that cause a specific joint movement

8. The .... joint allows us to turn our head to look from one side to the other.
   a. C1-C2
   b. C2-C3
   c. C3-C4
   d. C4-C5

9. Movement is more limited in the thoracic region of the spine because the ....
   a. Transverse processes are too long
   b. Spinous processes project straight out posteriorly
   c. Transverse processes angle backwards
   d. Vertebrae are connected to the ribs

10. How many spinal nerves are there in human body?
   a. 31
   b. 32
   c. 33
   d. 62

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Date: 14.06.11, Tuesday **Time: 10.20a.m to1.00p.m** **Maximum Marks: 60**

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II

SECTION –I

Q.1. Long Question (1 out of 2) **(1× 10= 10)**

1. What is a lever? Explain the various types of levers with examples and add a note on application of levers in physiotherapy. **(OR)**
2. What is function of skeletal system? Write down composition, structure and types of bones.

Q.2. Write short notes on: (4 out of 5) **(4× 5 = 20)**

1. Describe Newton's Laws of Motion with suitable examples.
2. What is equilibrium? Discuss the types of equilibrium with examples.
3. Types of Kinetic chains with examples in human body.
4. Explain the bucket handle and pump handle movement of rib cage.
5. Explain the role inter vertebral disc and its biomechanical significance.

SECTION –II

Q.3. Long Question (1 out of 2) **(1× 10= 10)**

1. Define Posture. Differentiate between Good and Bad Posture. List factors affecting Posture. **(OR)**
2. Define and classify Joints. Describe functions of Joints

Q.4. Write short notes on: (4 out of 5) **(4× 5 = 20)**

1. Phases of Respiration.
2. Discuss the various types of muscle contraction with examples.
3. Active & passive insufficiency.
4. Reflex Movement.
5. Mandibular Elevation and Depression.